

Multivariate statistics

Lecture 10: Classification with two populations

Literature and Problems: See course schedule

Agenda (preliminary):

Model

- populations and sample space
- decision rule and splitting of sample space
- **DEMO: two-population classification, case I and II**
- **SLIDE: Two-population classification, Training data vs. new data**
- possible misclassifications, confusion matrix and cost matrix, priors

General classification criteria (distribution free)

- ECM, Expected Cost of Misclassification, definition
- minimizing ECM ---> splitting of sample space ---> decision rule
- special case: equal costs ~ minimum TPM, Total Prob. of misclassification
- special case: equal costs and priors ~ ML-rule
- MAP, Maximum Aposteriori rule: Bayes rule, posteriors ---> MAP-rule

Classification for two multivariate normal (MVN) populations, case I (LDA)

- model with equal population covariance matrices
- LDA, Linear Discriminant Analysis decision rule (population version)
- sample based LDA decision rule
- alternative formulation for ML-case with squared statistical distances

MVN model control

- test for MVN samples
- Bartlett test for equal population covariance matrices
- Hotelling T^2 test for significant separation of populations

DEMO: two-population classification (LDA case)

Classification for two multivariate normal (MVN) populations, case II (QDA)

- model with unequal population covariance matrices
- QDA, Quadratic Discriminant Analysis decision rule (population version)
- sample based QDA decision rule

DEMO: two-population classification (QDA case)

Performance of classification ~ probability of misclassification

- TPM, analytical calculation, only theoretically possible
- Empirical APER, Apparent Error Rate, using training data confusion matrix
- Estimated AER, Actual Error Rate, using training data with cross-validation

DEMO: two-population classification, performance of LDA and QDA cases

SLIDE: Summarizing two-population classification procedures